Example 5.15

Find the length of Latus rectum of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Solution

The Latus rectum LL' (Fig. 5.22) of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through $S(ae,0)$.

Hence L is (ae, y_1).

Therefore,
$$\frac{a^2e^2}{a^2}+\frac{y_1^2}{b^2}=1$$
$$\frac{y_1^2}{b^2}=1-e^2$$
$$y_1^2=b^2(1-e^2)$$
$$=b^2\left(\frac{b^2}{a^2}\right) \quad \left(\text{since, } e^2=1-\frac{b^2}{a^2}\right)$$
$$y_1=\pm\frac{b^2}{a}.$$

That is, the end points of Latus rectum L and L' are $\left(ae,\frac{b^2}{a}\right)$ and $\left(ae,-\frac{b^2}{a}\right)$.

Hence the length of latus rectum $LL'=\frac{2b^2}{a}$.

(ii) Types of ellipses with centre at (h,k)

(a) Major axis parallel to the x-axis

From Fig. 5.24
$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1,\ a>b$$

The length of the major axis is $2a$. The length of the minor axis is $2b$. The coordinates of the vertices are $(h+a,k)$ and $(h-a,k)$, and the coordinates of the foci are $(h+c,k)$ and $(h-c,k)$ where $c^2=a^2-b^2$.

(b) Major axis parallel to the y-axis

From Fig. 5.25
$$\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1,\ a>b$$

The length of the major axis is $2a$. The length of the minor axis is $2b$. The coordinates of the vertices are $(h,k+a)$ and $(h,k-a)$, and the coordinates of the foci are $(h,k+c)$ and $(h,k-c)$, where $c^2=a^2-b^2$.

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